

Customer Shopping Behaviour Analysis

1. Project Overview

This project analyses customer shopping behaviour using transactional data from **3,900** purchases across multiple product categories. The objective is to uncover insights into spending patterns, customer segments, product preferences, and subscription behaviour in order to support data-driven business decisions.

2. Dataset Summary

- Number of Rows: 3,900
- Number of Columns: 18

Key Features:

- Customer demographics: Age, Gender, Location, Subscription Status
- Purchase details: Item Purchased, Category, Purchase Amount, Season, Size, Color
- Shopping behaviour: Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type

Missing Data:

- 37 missing values identified in the Review Rating column.

3. Exploratory Data Analysis using Python

The initial phase of the analysis focused on data preparation and cleaning using Python.

Data Preparation Steps:

- Data Loading: Imported the dataset using the pandas library.
- Initial Exploration: Examined dataset structure using `df.info()` and reviewed summary statistics with `df.describe()`

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

Previous Purchases	Payment Method	Frequency of Purchases
900.000000	3900	3900
NaN	6	7
NaN	PayPal	Every 3 Months
NaN	677	584
25.351538	NaN	NaN
14.447125	NaN	NaN
1.000000	NaN	NaN
13.000000	NaN	NaN
25.000000	NaN	NaN
38.000000	NaN	NaN
50.000000	NaN	NaN

- **Missing Data Handling:** Checked for null values and imputed missing values in the *Review Rating* column using the median rating of the corresponding product category.
- **Column Standardization:** Renamed all column names to *snake_case* to improve readability and maintain consistency.
- **Feature Engineering:**
 - a. Created an `age_group` column by binning customer ages.
 - b. Derived a `purchase_frequency_days` column based on purchase data.
- **Data Consistency Check:** Verified redundancy between `discount_applied` and `promo_code_used`. Since both conveyed similar information, the `promo_code_used` column was removed.
- **Database Integration:** Connected the Python environment to PostgreSQL and loaded the cleaned DataFrame into the database for further SQL-based analysis.

4. Data Analysis using SQL (Business Transactions)

Structured SQL queries were executed in PostgreSQL to answer key business-oriented questions:

1. **Revenue by Gender** – Compared total revenue generated by male and female customers.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. High-Spending Discount Users – Identified customers who used discounts yet spent more than the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	66

Total rows: 839 Query complete 00:00:

3. Top 5 Products by Rating – Retrieved products with the highest average review ratings.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. Shipping Type Comparison – Compared average purchase amounts between Standard and Express shipping methods.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. Subscribers vs. Non-Subscribers – Analysed differences in average spending and total revenue based on subscription status.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. Discount-Dependent Products – Identified five products with the highest percentage of discounted purchases.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

7. Customer Segmentation – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. Top 3 Products per Category – Listed the most frequently purchased products within each category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. Repeat Buyers & Subscriptions – Evaluated whether customers with more than five purchases were more likely to subscribe.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. Revenue by Age Group – Calculated the total revenue contribution of each age group.

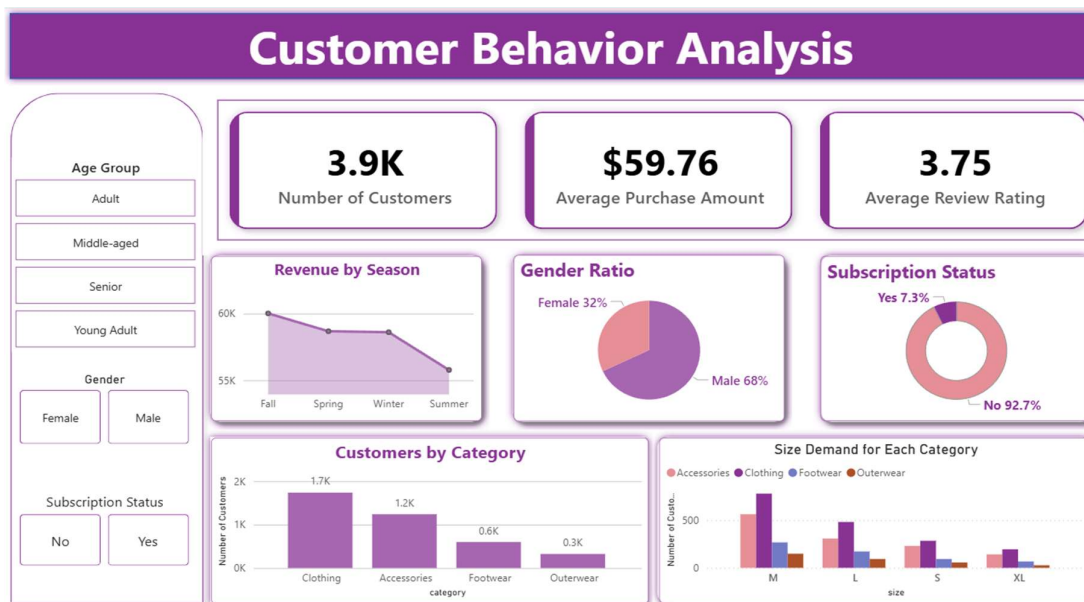
	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Dashboard in Power BI

An interactive Power BI dashboard was developed to visualize key insights from the analysis, enabling stakeholders to explore trends, compare segments, and monitor business performance effectively.

Sheet I

It represents the visual analysis of the data of customer shopping behaviour provided to us by the use of area chart, pie chart, donut chart and column charts as well.



Sheet II

It shows a pivot table representing the number of buyers (subscribers and non-subscribers) and average purchase amount of each item divided by category. A portion of it has been displayed here.

Total Revenue by Item Purchased, Category and Age group						
discount_applied age_group	No		Yes		Total	
	Count of customer_id	Avg. Purchase Amount	Count of customer_id	Avg. Purchase Amount	Count of customer_id	Avg. Purchase Amount
Young Adult	585	61.19	443	59.48	1028	60.45
Footwear	82	59.93	66	66.17	148	62.71
Boots	17	69.06	13	64.31	30	67.00
Shoes	16	59.56	19	70.16	35	65.31
Sandals	26	57.42	14	65.57	40	60.28
Sneakers	23	56.26	20	64.00	43	59.86
Accessories	173	62.32	145	58.40	318	60.53
Gloves	23	68.43	18	57.67	41	63.71
Sunglasses	26	58.54	15	69.33	41	62.49
Scarf	19	58.68	17	65.53	36	61.92
Backpack	20	66.50	17	55.94	37	61.65
Jewelry	21	67.38	19	52.84	40	60.48
Hat	19	58.37	20	61.05	39	59.74
Handbag	21	56.71	17	60.12	38	58.24
Belt	24	63.54	22	49.00	46	56.59
Clothing	280	60.81	199	59.16	479	60.13
Jeans	19	69.32	16	61.50	35	65.74
T-shirt	26	64.77	15	66.33	41	65.34
Dress	28	69.93	16	54.38	44	64.27
Shirt	29	61.41	24	65.08	53	63.08
Skirt	36	65.97	14	52.36	50	62.16
Shade	20	54.30	10	67.50	30	60.66
Total	2223	60.13	1677	59.28	3900	59.76

6. Business Recommendations

Based on the insights obtained from the analysis, the following recommendations are proposed:

- Boost Subscriptions: Promote exclusive benefits and incentives to encourage customer subscriptions.
- Customer Loyalty Programs: Implement reward mechanisms to convert repeat buyers into loyal customers.
- Review Discount Policy: Optimize discount strategies to balance increased sales with profit margin control.
- Product Positioning: Highlight top-rated and best-selling products in marketing campaigns.
- Targeted Marketing: Focus promotional efforts on high-revenue age groups and customers who prefer express shipping.